



A patient-centered approach to inclusive AI in healthcare: potential benefits, challenges, and recommendations

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Abstract

Purpose Artificial Intelligence (AI) has transformed various sectors, including healthcare, where it holds promises for improving early detection, personalized care, and decision-making. However, despite its potential, AI systems often exacerbate health inequalities, particularly among underrepresented communities, due to biased algorithms, limited accessibility, and data exclusion. This current study aims to examine inclusive AI in healthcare grounded on a patient-centered approach.

Methods We conducted a literature review following PRISMA 2020 guidelines, searching across online databases for publications from diverse geographic and cultural contexts. Articles were assessed for inclusion based on specific criteria, and data on healthcare, AI type, inclusivity frameworks, implementation methods, and SWOT analysis were extracted. The quality and risk of bias of included articles were assessed with the JBI Checklist for Expert Opinion; and the PROBAST tool.

Results Findings from the study suggest that various AI types were discussed, including machine learning, chatbots, and convolutional neural networks. Several key frameworks for ensuring inclusive AI were reported in the included articles, such as ACCEPT-AI, Bias Mitigation, and human rights values frameworks. However, some articles did not mention specific inclusivity frameworks. Our findings suggest that AI in healthcare has the potential to improve health outcomes but must be implemented inclusively and transparently to avoid exacerbating inequalities.

Conclusions This review emphasizes the need for ethical, regulated, and collaborative approaches to mitigate bias, enhance accessibility, and ensure patient-centered solutions. Future efforts should focus on stakeholder-inclusive design and standardized frameworks for equitable AI in healthcare.

Registration This literature review was not registered.

Keywords Artificial intelligence · Algorithms · Health · Health inequalities · Inclusive AI

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Abbreviations

AI	Artificial Intelligence
CNNs	Convolutional neural networks
LLMs	Large Language Models
SWOT	Strengths, Weaknesses, Opportunities, and Threats

1 Introduction

Artificial Intelligence (AI), which are computer systems that have the capability to imitate human thinking, is transforming society e.g., from self-driving cars, facial recognition, fraud detection, autonomous systems, etc [1]. Indeed, the integration of AI into society has substantially improved human activities, highlighting its profound impact on our day-to-day experiences paving the way for more effective

decision-making in different sectors and communities [2]. Similarly, over the last decade AI is now being deployed in the healthcare sector in various fields of medicine such as clinical decision support, drug discovery, medical imaging analysis prognostication of diseases or disease outcomes, protein modeling, personalized medicine, etc [1, 3]. Advances in AI technologies hold great promise for improving early screening and detection methods, and improving or informing personalized healthcare outcomes [4]. These technologies have the potential to analyze large, multidimensional datasets spanning user, clinical and other types of data in ways that enable informed decision-making, targeted interventions and precision care [5].

Irrespective of the huge potential of AI in healthcare its use has paradoxically increased health inequalities and exclusion especially amongst unrepresented communities, race, gender, and ethnic minorities [6]. This is because AI algorithms and models that analyze large amounts of user- or clinically-generated health data tend to exclude data from certain user populations [6]. Moreover, there are growing concerns that AI systems may further exacerbate inequalities in healthcare due to inconsistent accessibility, the lack of affordability, and biased AI algorithms and models [5]. This is in contrast to an inclusive society that offers equal opportunities for all individuals irrespective of their age, race, gender, or social class [7, 8]. Inclusion refers to the degree to which individuals feel integrated or subjectively accepted within society. In the context of this study, an inclusive healthcare system ensures equitable access for all [8]. Inclusive design focusses on developing services, environments, and products that accommodate individuals with diverse characteristics, needs, and abilities [9].

In the emerging era of digitalization and AI in healthcare, this means that digital health solutions, including AI-based health technologies, need to be designed with a patient-centered approach, so that users from different backgrounds and with diverse needs can interact with them seamlessly and benefit equally. Thus, it is essential to explore strategies for bridging the technological divide, particularly in underserved regions, to address disparities faced by patients experiencing social disadvantages and technological barriers [5, 10]. Equally important is the development of inclusive AI systems (e.g. AI systems designed to mitigate disparities across gender, race, and socioeconomic status) tailored to diverse patient groups, considering factors such as language discordance between healthcare providers and patients to mitigate technological exclusion in healthcare [10]. This includes developing AI systems that promote better understanding and trust between patients and healthcare providers [10]. The ethical guidance developed by the World Health Organization in 2021 emphasizes the importance

of inclusivity, transparency, and accountability in AI for healthcare [11].

An increasing number of published reviews on the use of AI in healthcare focus primarily on the technological advancements and algorithmic performance on various fields, such as cardiology, dentistry, surgery, disease surveillance, rehabilitation, or precision medicine among others [12–17]. However, many of these AI systems, despite their technical sophistication, have been shown to perpetuate or even exacerbate health disparities due to the use of biased data or to design processes that overlook underrepresented groups. For example, several AI systems for dermatology have demonstrated significant racial bias [18, 19]. There is also an ongoing research on how to mitigate biases in AI systems and address health disparities in healthcare through stakeholder involvement and human-centered AI principles [20]. Other studies emphasize the need for diverse datasets, fairness-conscious algorithms, and regulatory frameworks for inclusive AI. To the best of our knowledge, an overview on inclusive AI and strategies and frameworks to achieve “inclusivity in AI” in healthcare is still missing. Where inclusivity in AI refers to the development and implementation of AI systems or models that are fair, accessible, and useful to all, irrespective of gender, background, race, or ability. In the context of this study inclusivity in AI aims to address or reduce algorithmic bias that may exist in diverse datasets. Thereby ensuring equitable access to AI systems or models, as well as the inclusion of diverse teams during the development of these AI systems or models.

As suggested by the World Health Organization [11], who emphasizes the importance of inclusivity, transparency, and accountability in AI for healthcare [11]. Our literature review aims to provide an overview of scientific research on the application of inclusive AI technologies in healthcare, highlighting their reported strengths, weaknesses, opportunities and threats. Our study was guided by the following research questions:

- What are the facets of inclusive AI, and what are the recommended strategies, solutions, and regulatory frameworks needed to realize inclusive AI technologies in healthcare?
- What are the application areas where inclusive AI technologies have been investigated?
- What are the potential benefits (strengths and benefits) of inclusive AI technologies within the healthcare domain?
- What are the potential challenges (weaknesses and threats) of inclusive AI technologies within the healthcare domain?

Overall, in summary this study contributes to the existing body of knowledge by examining the application areas of

inclusive AI technologies in healthcare, assessing their benefits and challenges, exploring the facets of inclusivity in AI, and most importantly, summarizing the proposed strategies, solutions and regulatory frameworks needed to achieve truly inclusive AI in healthcare. An additional objective of this study is to shape the AI research agenda for improving health inclusion when deploying AI systems among patients from diverse and disadvantaged communities. Evidence from this study highlights the intersection of inclusive AI with systemic bias and healthcare disparities, examining how AI-driven inequities affect low-income groups, people with disabilities, communities of color, minorities, and other disadvantaged populations.

2 Methods

In this study, a literature review was adopted as the chosen research approach by carrying out a systematic review with conceptual synthesis, following the PRISMA 2020 guidelines [21], as seen in Appendix 3. We searched six databases (PubMed, PsycINFO, CINAHL, Academic Search Premier, IEEE Xplore, and Scopus) for publications that included the keyword “Inclusive AI” in their abstract. In this study we prioritize the literature to explicitly addressing “inclusive AI” in healthcare as this area has not been well addressed in prior studies. Accordingly, we used only the keyword “Inclusive AI” “strategies”, “solutions”, “regulatory frameworks”, “applications”, “benefits”, “challenges”, AND “bias”, “fairness”, “equity”, OR “diversity” AND “healthcare” to ensure the search captured publications where inclusivity was an explicit and central focus. The search was conducted on the 26th of September 2024. The search was carried out again on the 16th of April 2026 after extending the search keyword based on the research questions being examined in this study to improve the literature. The full search strategy can be found in Appendix 1. We uploaded all identified references to EndNote, removed duplicates, and then uploaded the remaining references to Rayyan.

2.1 Eligibility and selection process

In order to assess eligibility, two authors (EG and KD) independently screened the titles and abstracts of all identified articles. In the second step, two authors (EG and AJB) independently assessed the eligibility of the selected articles from the first step by reviewing their full text. Any disagreements were discussed with a third author (KD) until consensus was reached.

We considered the following eligibility criteria:

Inclusion criteria:

- Articles addressing inclusive AI in the context of healthcare.
- Written in any language and published in any year.

Exclusion criteria:

- Publications for which only the abstract is available (full text not available).
- Study protocols, reviews, or protocols of systematic reviews.
- Editorials, letters to editor, erratums, corrigenda, correction, comments, retracted articles, responses (Re:), proceedings, books, etc.

2.2 Data items and data extraction

All selected articles were included in qualitative synthesis and were divided between two authors (EG and KD), with each author extracting data from half of the articles. After which further discussion was carried out among the three authors during data extraction. The following information was extracted from each study: Healthcare context (e.g., condition, medical specialty); Type of AI solution (e.g., AI-based prediction algorithm, natural language processing, chatbot); Use of frameworks and/or guidelines to ensure inclusivity in AI; Specific methods for implementing inclusivity within the AI solution; and the discussed SWOT data addressing the strengths, weaknesses, opportunities, and threats of inclusive AI solutions. The classification of discussed topics into SWOT categories was based on the characteristics highlighted in each article. When the categorization was explicitly stated by the authors, it was directly adopted. When not clearly reported, positive characteristics already in place and described positively were classified as Strengths, while positive characteristics expected to occur were classified as Opportunities. Negative characteristics that have already occurred were classified as Weaknesses, and negative characteristics anticipated to occur were classified as Threats. These classifications were performed by the author responsible for extracting the data.

From the results, we derived facets of inclusive AI solutions in healthcare, formulated recommendations addressing these facets to achieve inclusive AI in healthcare and formulated a research agenda.

2.3 Critical appraisal and risk of bias assessment

We conducted a critical appraisal and risk of bias assessment of all included articles. Viewpoints and expert opinion papers were assessed using the JBI Critical Appraisal Checklist for Expert Opinion, and articles involving the

development or testing of AI/ML models were appraised using the PROBAST tool.

3 Results

3.1 Articles selection

The search resulted in 1,377 results. After removing duplicates in EndNote and Rayyan, we kept 1,163 results for further screening. In the title and abstract screening, both reviewers agreed to exclude 1,124 references; 39 articles

were considered appropriate to be screened in depth during full-text review. Twenty-one were excluded at this stage; the list of excluded articles during the full-text screening and the reason for their exclusion is presented in Appendix 2. A total of 18 articles were included in this review [1, 5, 6, 8, 9, 22–34] (see Fig. 1, and Table 1). All included articles were published between 2020 and 2024.

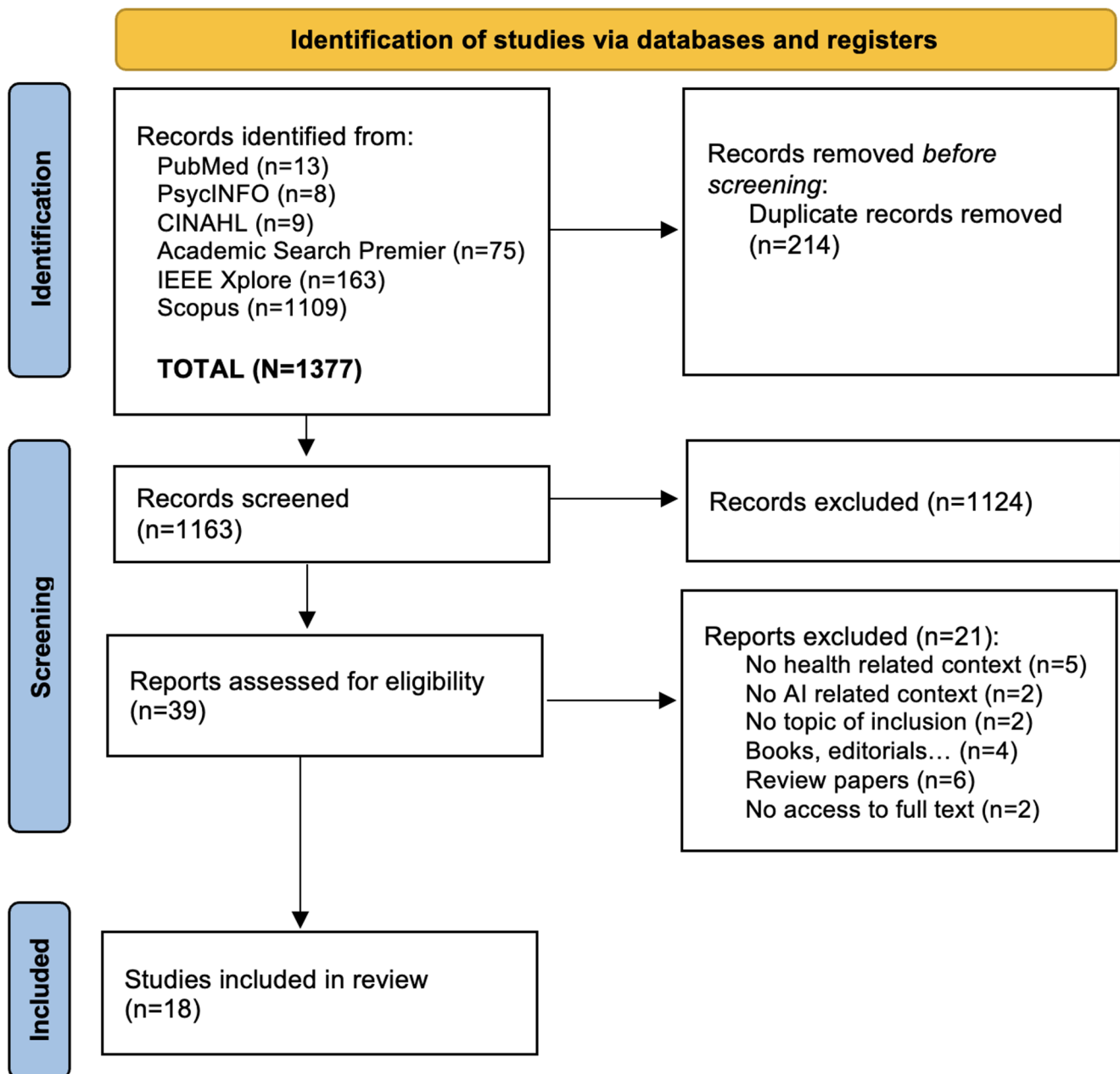


Fig. 1 Flowchart of the selection procedure

Table 1 Overview of the information extracted from the included publications ($n = 18$)

Reference	Healthcare context	AI solution	Used Frameworks	Strengths	Weaknesses	Opportunities	Threats
Alami et al. [22]	Healthcare in general	AI in general	Sustainable Development Goal 17 of the United Nations (“Partnership for the Goals”)	• Cost reduction • Resource optimization	• Lack of data • Risk of bias • Governance challenges • Quality and safety concerns • Dependence on external support • Limited follow-up care • Potential erosion of skills (may weaken clinical, critical thinking, and community health practice skills)	• Improved accessibility • Support for stigmatized populations • Cultural and linguistic adaptation • Epidemiological insights • Maternal and child health	• Adverse social impact • Data misuse • Unintended misuse • North-South divide • Over-medicalization • Marginalized populations at risk
Amugongo et al. [6]	Healthcare in general	AI in general	Ubuntu philosophy	• Fairness and inclusivity • Ubuntu principles (empathy, inclusivity, mutual support, ethical values) • Humanity-focused safeguards • Trust-building measures	• Data bias issues • Opacity of AI systems • Challenges with informed consent	• Improving fairness in healthcare • Building community trust • New regulatory frameworks • Enhancing personalization and empowerment • Collaboration potential between all stakeholders	• Risk of exacerbating inequalities • Ethical misuse of AI • Trust deficit • Accountability gaps
Avellan et al. [23]	Healthcare in general	AI in general	Not reported	• Diversity in algorithm design • Representative training data • Accessibility • Collaborative frameworks	• Bias in current AI systems • Challenges in ethical data collection • Accessibility barriers	• Collaborative development • Innovations in accessibility • Policy and ethical development	• Systemic bias and inequality • Regulatory and privacy challenges • Technological exclusion
Dankwa-Mullan et al. [5]	Cancer	AI and machine learning	Principles of distributive justice	• Improvements in diagnosis, prognosis, treatment and patient care • Diverse applications of AI • Incorporation of multi-omics data • Use of AI to reduce bias • Emphasis on data diversity • Focus on cancer health equity	• Bias in AI models • Underrepresentation in clinical trials • Challenges in data diversity • Over-reliance on historical data • Complexity of integrating diverse data sources • Biases in clinical care	• Enhanced precision medicine • Increased diversity in clinical trials • Improving data transparency and inclusivity • Collaboration with diverse stakeholders • AI role in decision-making • Integration of social determinants of health	• Reinforced bias in training data • Disparities in cancer care access • Risk of underrepresenting minority populations • Lack of universally accepted standards • Reliance on historical data • Possible exclusion of marginalized groups

Table 1 (continued)

Reference	Healthcare context	AI solution	Used Frameworks	Strengths	Weaknesses	Opportunities	Threats
Dekker et al. [24]	Mental health among students	Chatbot	Braun and Matthes framework for chatbot classification and several technology acceptance theories	• Scalable solution • Low threshold for access • Inclusive approach • Personalized and interactive coaching • Early detection • Higher adherence rates	• Limited human interaction • Potential over-reliance on technology • Lack of comprehensive research • Challenge in identifying specific student needs	• Expanding to a broader audience • Need to explore differentiated approaches • Potential to be integrated into wider curricula at universities	• Stigma around mental health • Technology acceptance and usability issues • Over-simplification of complex issues • Privacy and security concerns
Denecke et al. [9]	Radiology	AI and machine learning	Just-in-Time Adaptive Intervention design	• Emphasis on inclusivity • Collaboration across disciplines • Legislative support • Customizability	• Limited evaluation on content understanding • Lack of metrics for inclusivity • Preliminary research only	• Incorporation of cultural competence • Development of taxonomy for inclusive design • Visual aids and simplified content	• Underrepresentation of cognitive diversity • Limited tools for evaluation • Potential for misalignment with real-world needs
Doo et al. [27]	Radiology	AI and machine learning	“Digital determinants of health” sociotechnical framework	• Diverse stakeholder inclusion • Increased focus on diversity, equity and inclusion • Workforce integration • Commitment to training and education • Collaborative programs	• Lack of diversity, equity and inclusion in development and implementation • Historical and structural barriers • Gaps in AI education • Bias in data sets	• Inclusive design • Development of collaborative programs to train AI in radiology • Broadening access to AI to lower-resources populations • Improvement of patient care	• Exclusion of marginalized groups • Bias perpetuation • Workforce displacement • Unintended harms
Ekellem et al. [28]	Dermatology	CNNs	GUI (Graphical User Interface) principles	• potential to improve early disease detection • Validation of existing knowledge • Focus on racial equity	• Data bias regarding diverse skin • Poor performance in certain diagnoses • Limitations in current datasets	• Data collection and collaboration • Model refinement • Bias detection and mitigation • Continuous validation	• Reinforcing of systemic biases • Real-world consequences of biases models • Ethical and social justice concerns
Garcia-Saiso et al. [29]	Cancer	AI in general	Not reported	• Potential to enhance diagnostic precision, resource allocation, and make health more accessible • Improved cancer care outcomes • Potential for a more inclusive healthcare solution • AI-enhanced patient engagement • Educational tools	• Biased AI models • Data diversity gaps • Ethical concerns (privacy and transparency) • Cultural insensitivity • Limited adoption in low resource areas	• Potential to enhance cancer care across the globe • Can bridge geographical gaps • Personalized healthcare • Research collaboration • Address health inequities • Efficient pandemic response	• Widening health disparities • Resistance to adopt AI • Ethical misuse • Regulatory and policy challenges • Lack of infrastructure

Table 1 (continued)

Reference	Healthcare context	AI solution	Used Frameworks	Strengths	Weaknesses	Opportunities	Threats
Garett et al. [30]	HIV	AI in general	Not reported	• Willingness to adopt AI by physicians • Expert involvement • Reduced privacy risks • AI testing for credibility and trust	• Users mistrust • AI limitations (emotional support) • Risk of misleading information • Biased AI models • Omission of sensitive data may reduce accuracy of prediction	• Inclusive and participatory approaches • Federated learning for protecting patient information • AI for marginalized groups • Use of differential privacy • AI potential to aid in HIV prevention, surveillance and tailored care	• Potential to exacerbate stigma • Data privacy • Overload of information • Algorithm bias and discrimination
Griffin et al. [31]	Healthcare in general	AI in general	FAIR, FACT, and RE-AIM frameworks	• Ability to reduce health disparities through equitable AI implementation • Application of explainable AI and frameworks for transparency	• Lack of standardization • Biased intervention • Biases in data and algorithms, challenges with ensuring model transparency in complex AI systems • Uncertainties in how data is collected, issues of trust and adoption among clinicians and patients	• Using ethnographic methods to identify sociotechnical biases • Advancing data quality through improved granularity • Promoting diverse team involvement in AI development	• Risks of exacerbating disparities if biases in AI systems remain unaddressed • Potential for over-reliance on incomplete or biased datasets • Ethical concerns
Iloanusi et al. [1]	Healthcare in general	AI and machine learning	Not reported	• Expanding access to care • Enable personalized medicine • Augment clinical decisions • Analyze diverse health data • Promote system equity • Can identify social determinants of health disparities	• Historical discrimination due to reliance on historical data • Algorithmic bias • Lack of diverse data • Ethics • Uncritical trust in AI's objectivity	• Ensuring participatory design processes • Building governance structures incentivizing equity-promoting AI • Evolving regulatory and accountability mechanisms to detect and prevent algorithmic harm	• Potential misuse of AI systems to exacerbate discrimination or limit care for vulnerable population • Challenges in addressing power dynamics • Profit motives, and lack of incentives for equity-focused AI • Ethical concerns around automation replacing human judgement • Limited regulatory mechanisms

Table 1 (continued)

Reference	Healthcare context	AI solution	Used Frameworks	Strengths	Weaknesses	Opportunities	Threats
Israni et al. [25]	Adolescent and young adult health	AI in general	Human rights values framework	• Personalization • Interdisciplinary approach • Potential to advance healthcare	• Historically biased data • Lacking clear consent for data usage • Privacy concerns • Digital divide	• Advancing preventive care (improve early interventions) • Potential to mitigate the digital divide through inclusive technology design • Regulatory improvements	• Psychological and social risks (vulnerabilities from technological misuse, such as cyberbullying and identity exploitation) • Misaligned objectives: Business-centric AI deployment may prioritize efficiency over care
Kalina [8]	Access to healthcare services	AI in general	Bias Mitigation Framework	• Potential to improve health equity • Ability to process vast and diverse datasets could lead to precision medicine	• Quality of and bias in healthcare data, ignoring minorities in population • Reliance on technology • Lack of robust frameworks for mitigating biases and ensuring equitable outcomes	• Advancing health equity by eliminating existing healthcare access barriers • Enhancing patient engagement and trust through participative and transparent AI development	• Risk of exacerbating existing health disparities • Ethical dilemmas and patient distrust in AI-based decisions • Challenges in achieving international standards for inclusive AI
Maheshwari, et al. [32]	Newborn screening program	AI and machine learning	Framework of institutional epistemic trust	• Potential for accurate and timely detection of genetic conditions • Reduction of manual labor and increased efficiency • Support for early interventions, enhancing specific medical tasks	• Risks of opacity and black-box problem in AI-decision making • Burden on healthcare professionals to understand and communicate AI-generated insights • Increased dependence on digital and health literacy among patients • Potential for trust undermining if ethical or legal frameworks are not adhered to	• Development of AI literacy through education and public awareness campaigns • Creation of algorithm registers and transparent practices to improve trust • Leveraging AI for expanding screening capabilities to detect more conditions	• Public mistrust originating from unfamiliarity with AI • Ethical concerns over data privacy and commercial exploitation of health data • Regulatory and societal challenges in ensuring AI's equitable integration • Risks of over-reliance on systems

Table 1 (continued)

Reference	Healthcare context	AI solution	Used Frameworks	Strengths	Weaknesses	Opportunities	Threats
Muralidharan, et al. [33]	Global child health with emphasis on LMICs	AI and machine learning	ACCEPT-AI, SPIRIT-AI, and CONSORT-AI	• Focus on age-specific challenges in pediatric AI/ML • Advocacy for global collaboration	• Limited data related to pediatrics, data quality, data variety, ethical and regulatory frameworks for pediatric AI/ML • Over-reliance on datasets from high-income countries leading to potential biases	• Leveraging AI/ML to address critical pediatric health challenges like malaria, tuberculosis, and malnutrition • Development of international pediatric data-sharing networks to create representative training datasets • Enhanced focus on co-design methodologies involving communities and stakeholders globally	• Risk of exacerbating global health inequities if LMICs are excluded from AI advancements • Challenges in data privacy and governance for pediatric populations • Potential harm from poorly implemented AI tools
Rubeis et al. [26]	Active ageing	AI and machine learning	Social Justice Framework for Bridging the Digital Divide and Health Equity Impact Assessment	• Enables agency and social participation, enhance independent living, well-being, and personalized care for older adults • Opportunities to reduce inequalities by integrating social determinants into design and data collection • Encourages participatory design	• Bias in training data • Oversimplification • Risk of stereotypical narratives • Access barriers due to low digital literacy • Challenges in managing complexity	• Enhanced global collaboration to adapt AgeTech for diverse cultural and economic contexts • Empowering older adults with e-health literacy programs	• Socioeconomic disparities in access to advanced solutions • Ethical dilemmas arising from over-standardization and potential user conditioning
Tielman et al. [34]	People with cognitive disabilities	Adaptive and personalized explanation systems for AI interactions	IEEE 7000 standard for ethical system design, W3C Accessibility Guidelines, and a socio-cognitive engineering method	• Personalization and tailoring explanations to individual needs • Use of interdisciplinary approaches to bridge AI development and user requirements • Inclusion of moral and ethical considerations in system design	• Lack of coherent set of design guidelines • Lack of real-world studies • Challenges in integrating user feedback into AI system design at scale • Limited availability of datasets representing users with cognitive disabilities	• Participatory design for more effective and inclusive AI solutions • Expanding research into multimodal and adaptive explanation mechanisms	• Risk of oversimplification leading to loss of critical information • Potential exclusion of specific cognitive profiles due to one-size-fits all approach • Ethical concerns over misuse of personal data collected for tailoring explanations

3.2 Critical appraisal of the included articles

All 17 expert opinion articles met the JBI Checklist criteria for expert opinion, with unanimous “Yes” ratings across the first five domains: clear identification of the opinion

source, author standing in the field of expertise, central focus on the interests of the relevant population (marginalized, racial, and ethnic minorities in AI-health contexts), logically defended arguments supporting the conclusions drawn, and comprehensive reference to the extant literature

(See Table 2). For the sixth domain, 10 articles reported no significant incongruence with existing sources, while the remaining seven explicitly acknowledged and logically defended noted incongruences, including the dominance of high-income country perspectives requiring adaptation to low- and middle-income settings, the integration of non-Western ethical frameworks such as Ubuntu, the limitations of individual-responsibility models in distributed AI systems, gaps in the literature, the generally optimistic framing of AI and AgeTech, and author's optimism despite current harms associated with AI literature (See Table 2).

Using PROBAST, the study of Ekellem et al. [28] was judged to have a high risk of bias across specially in the analysis domain due to overfitting, lack of proper validation, and small sample size. Additional concerns were identified in participant selection and outcome definition, limiting the generalizability and reliability of the findings (See Table 3).

3.3 Context: healthcare domains and types of AI solutions

Regarding the healthcare context of AI use, six articles referred to AI for healthcare in general [1, 6, 9, 22, 23, 31]; while the rest focused on specific topics, such as AI for adolescent and young adult health [25], AI in global child health with an emphasis on low- and middle-income countries [33], in newborn screening programs [32], for active aging [26], as a means of accessing healthcare services [8], in cancer care [5, 29], radiology [27], dermatology [28], in HIV applications [30], for mental health among students [24], and for people with cognitive disabilities [34].

Eight articles discussed AI in general [6, 8, 22, 23, 25, 29–31], seven AI and machine learning [1, 5, 9, 26, 27, 32, 33], one focused on adaptive and personalized explanation systems for AI interactions [34], one discussed a chatbot [24], and one article referred to convolutional neural networks (CNNs) in AI applications [28].

3.4 Frameworks to ensure inclusivity, methods and definitions of inclusive AI

Several frameworks and guidelines supporting inclusive AI were reported in the included articles. One article referred to the ACCEPT-AI, SPIRIT-AI, and CONSORT-AI frameworks [33]. Another article discussed a Bias Mitigation Framework and ethical AI design principles [8], while one highlighted the FAIR, FACT, and RE-AIM frameworks [31]. Maheshwari et al. referred to a framework of institutional epistemic trust [32], and Israni et al. mentioned a human rights values framework [25]. Tielman et al. outlined

the IEEE 7000 standard for ethical system design, W3C Accessibility Guidelines, and a socio-cognitive engineering method [34]. Doo and McGinty referred to the “digital determinants of health” sociotechnical framework [27], while Dekker et al., mentioned the Braun and Matthes framework for chatbot classification and several technology acceptance theories [24]. Ekellem and Köhler referred to GUI (Graphical User Interface) principles [28], and Denecke et al. discussed Just-in-Time Adaptive Intervention design [9]. One article highlighted the principles of distributive justice [5], and another one referred to Sustainable Development Goal 17 of the United Nations (“Partnership for the Goals”) [22]. Rubeis et al. mentioned the Social Justice Framework for Bridging the Digital Divide and Health Equity Impact Assessment [26], while one article discussed the Ubuntu philosophy [6]. Four articles [1, 23, 29, 30] did not mention any specific framework or guideline for ensuring inclusive AI.

3.5 Strengths, weaknesses, opportunities and threats

The identified strengths, weaknesses, opportunities and threats are listed in Table 1 and can be grouped as shown in Fig. 2. The SWOT categorization was derived from the literature which was later assessed by all authors to ensure consistency of the findings. Inclusive AI fosters accessibility and personalized healthcare solutions by tailoring services to individual needs while ensuring equity in healthcare delivery. Additionally, patient engagement is strengthened through interactive tools and AI-powered coaching that provide personalized guidance. A key strength of inclusive AI is its ability to integrate diversity at all levels, from algorithm design to data representation. The use of diverse, representative training datasets helps reduce bias and ensure AI solutions work effectively across varied demographics. Building trust in AI systems requires transparency and ethical design. Humanity-focused safeguards help mitigate bias while ensuring that AI applications remain explainable and fair. AI-driven automation lowers healthcare costs while improving service delivery by enabling early interventions and enhancing clinical decision-making. Improved accessibility and inclusivity was one of the most commonly discussed strengths, with 12 of the 18 included articles highlighting how AI enables personalized medicine, early detection, and enhanced patient engagement to address diverse needs and promote equity [5, 8, 9, 23, 24, 26–29, 31, 32, 34].

We extracted several threats from the papers. Bias in AI algorithms, underrepresentation of marginalized groups, and exclusion of low-income regions contribute to unequal healthcare access and outcomes. These disparities can

Table 2 Critical appraisal based on JBI checklist for expert opinion

Reference	Is the source of the opinion clearly identified?	Does the source of opinion have standing in the field of expertise?	Are the interests of the relevant population the central focus of the opinion?	Does the opinion demonstrate a logically defended argument to support the conclusions drawn?	Is there reference to the extant literature?	Is any incongruence with the literature/sources logically defended?
Alami et al. [22]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged (e.g., dominant literature on AI's benefits, but largely from high-income countries and needs adaptation to LMICs)
Amugongo et al. [6]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged and defended (e.g., Ubuntu principles overlap Western AI ethics)
Avellan et al. [23]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged (e.g., most frameworks focus on individual human responsibility rather than distributed agency in AI systems)
Dankwa-Mullan et al. [5]	Yes	Yes	Yes	Yes	Yes	No incongruence
Dekker et al. [24]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged (e.g., limitations and gaps in literature)
Denecke et al. [9]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Doo et al. [27]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Garcia-Saiso et al. [29]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Garett et al. [30]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Griffin et al. [31]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Iloanusi et al. [1]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Israni et al. [25]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged (e.g., authors do not dispute the promise of the optimistic literature on AI's potential)
Kalina, 2023 [8]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged (e.g., author's optimism despite AI's current harms)
Maheshwari, et al. [32]	Yes	Yes	Yes	Yes	Yes	No significant incongruence
Muralidharan, et al. [33]	Yes	Yes	Yes	Yes	Yes	No significant incongruence

Table 2 (continued)

Reference	Is the source of the opinion clearly identified?	Does the source of opinion have standing in the field of expertise?	Are the interests of the relevant population the central focus of the opinion?	Does the opinion demonstrate a logically defended argument to support the conclusions drawn?	Is there reference to the extant literature?	Is any incongruence with the literature/sources logically defended?
Rubeis et al. [26]	Yes	Yes	Yes	Yes	Yes	Yes, incongruence acknowledged (e.g., authors do not dispute the optimistic literature on AgeTech's benefits)
Tielman et al. [34]	Yes	Yes	Yes	Yes	Yes	No significant incongruence

Table 3 Risk of bias based on PROBAST

Reference	Participants	Predictors	Outcomes	Analysis	Overall
Ekellem et al. [28]	High	Unclear	High	High	High

reinforce existing inequities, leading to poorer health results for disadvantaged populations. AI-driven healthcare faces challenges related to data privacy, commercial exploitation, and the absence of clear consent mechanisms. Misuse of patient data to train AI models raises ethical and legal concerns, potentially eroding trust in AI applications. Over-reliance on AI technologies may lead to diminished clinical

skills among healthcare professionals and oversimplification of complex medical decision-making. Inadequate infrastructure, governance gaps, and global disparities hinder the seamless adoption of AI in healthcare. Resource-poor regions may struggle to implement AI solutions effectively, exacerbating existing inequalities. Resistance to AI adoption stems from fears of AI replacing human judgement, ethical dilemmas, and a lack of public trust. Concerns about accountability, transparency and fairness further slow the adoption of AI in medical decision making. From these threats, the most commonly discussed threat identified from the selected and reviewed sources was the exacerbation of inequities, noted in 14 articles, warning that biases

STRENGTHS Enhancing accessibility, inclusivity, personalized medicine, early detection, patient engagement and promoting equity [5,8,9,23,24,26–29,31,32,34] Diversity in data representation and AI development [5,9,23,25,27,30,33,34] Promotion of global collaboration and co-design, accessibility, and scalability [1,9,23,24,26] Promotion of transparency, trust, and bias reduction through humanity-focused principles [5,6,30,31,34] Improvement of diagnostic precision and support of multi-omics integration for precision medicine [1,5,8,29] Optimization of resources; cost reduction, improvement of healthcare delivery [1,22,32]	WEAKNESSES Underrepresentation in datasets and reliance on biased historical data limit inclusivity and fairness [1,5,6,8,9,22,23,25–31,33,34] Lack of robust frameworks, clear consent mechanisms, and real-world evaluations undermines trust and adoption [1,6,8,9,24,25,30–32,34] Challenges in managing AI system opacity and integrating user feedback at scale [6,22,23,29,31,32] Issues with cultural sensitivity, privacy, and over-reliance on AI for critical decision-making [5,8,22,24,29] Exacerbating inequities through barriers such as low digital literacy and limited access in resource-poor settings [23,26,29,32]
OPPORTUNITIES Potential to bridge disparities by integrating social determinants of health and improving access for marginalized populations [5,9,22,24,27,28,30,33] Fostering trust and transparency by inclusive design frameworks, regulatory advancements, and participatory approaches [1,6,8,25,27,28,31,32] Expanding data diversity and applicability through enhancement of co-design methodologies and international partnerships [5,6,23,26,33,34] Refinement of models and enhancing personalization to advance in data granularity and federated learning [9,25,28–31] Development of AI literacy for patients, providers, and policymakers to increase adoption and reduce mistrust [8,26,27,32]	THREATS Widening of health disparities because of bias in algorithms, underrepresentation and exclusion of low-income regions [5,6,8,9,22,23,26–31,33,34] Significant challenges caused by data misuse, commercial exploitation, and lack of consent mechanisms [1,22,25,29–32,34] Skill erosion and oversimplification of complex healthcare dynamics through dependence on AI technologies [1,24–27,32–34] Limited equitable integration through insufficient infrastructure, governance gaps, and global disparities [1,5,6,8,23,29,33] Lack of adoption through public skepticism and ethical concerns about AI replacing human judgment [8,24,25,32]

Fig. 2 SWOT Analysis of commonly discussed topics in the literature on inclusive AI in healthcare, with references to the articles reporting each aspect

in algorithms and exclusion of low-income regions could further widen health disparities [5, 6, 8, 9, 22, 23, 26–31, 33, 34].

Many AI models inherit biases due to the underrepresentation of diverse populations in datasets and reliance on historically biased data. This leads to unfair and less inclusive outcomes. Aspects contributing to these weaknesses mentioned in the papers are lack of diverse data, over-reliance on datasets from high-income countries or bias in clinical care. Another weakness is the lack of standardized frameworks, ethical guidelines, and clear user consent processes. This hinders AI adoption by mistrust and concerns about fairness. AI decision-making is often a “black box,” making it difficult to understand and correct biases. Lack of user input integration further limits inclusivity. AI systems can perpetuate cultural insensitivity, breach privacy, and lead to excessive dependence on technology for essential decision-making. Disparities in access to AI-driven tools such as accessibility barriers or limited adoption in low-resource areas create gaps in inclusivity, particularly for underserved populations. The most frequently mentioned weaknesses were bias and data limitations, cited in 16 articles, emphasizing how underrepresentation in datasets and reliance on biased historical data limit inclusivity and fairness [1, 5, 6, 8, 9, 22, 23, 25–31, 33, 34].

We also identified several opportunities for inclusive AI. Inclusive AI can play a transformative role in reducing healthcare and digital inequities by improving access for marginalized populations and addressing social determinants of health. By applying inclusive design frameworks, considering regulatory frameworks and participatory approaches during the development, inclusive AI can foster trust and transparency of AI solutions. Co-design methodologies and international collaborations in aggregating data sets to be used in AI systems may improve AI fairness. Educating stakeholders on AI applications and limitations can improve adoption and reduce skepticism. AI can improve in accuracy and personalization by refining models, improving data collection, and leveraging privacy-preserving learning techniques. Among opportunities, we identified two big topics as the most commonly discussed: the potential in advancing health equity and innovations in design [5, 9, 22, 24, 27, 28, 30, 33] and the need of inclusive frameworks and policy [1, 6, 8, 25, 27, 28, 31, 32], each discussed in 8 articles.

4 Discussion

4.1 Summary of findings

A total of 18 articles were included in this literature review, all published between 2020 and 2024. The critical appraisal indicated that the majority of articles included were expert opinion papers meeting key JBI quality criteria, suggesting a generally strong conceptual and theoretical foundation. However, the predominance of expert opinion highlights a limited availability of empirical evidence in the field of inclusive AI in healthcare. These articles focused on AI applications in health, including general healthcare, cancer care, mental health, and specific conditions like cognitive disabilities. These articles referred to various AI types, including machine learning, chatbots, and convolutional neural networks. Multiple frameworks for ensuring inclusive AI were reported, such as ACCEPT-AI, Bias Mitigation, and human rights values frameworks. However, some articles did not mention specific inclusivity frameworks. The SWOT analysis highlights improved accessibility and inclusivity as a key strength, while the most frequently identified weakness is bias and data limitations. Among the opportunities, advancing health equity and fostering innovation in design, frameworks, and policy were the most discussed topics. The most commonly cited threat was the potential for AI to exacerbate inequities, excluding some race, certain gender, thereby widening health disparities. We identified several recurring themes related to the participatory health approach that AI technologies are supporting. These themes center on empowering patients, improving communication and relationships, promoting shared decision-making, and fostering equitable healthcare practices. These application areas support the need to design AI technologies inclusively in order not to exclude individuals from benefiting from healthcare services delivered by AI technologies [4].

4.2 Facets of inclusive AI in healthcare

Figure 3 depicts the facets of inclusive AI in healthcare identified from the literature [35–38]. The identified facets of inclusive AI in healthcare as they can be concluded from the SWOT analysis results, reflect its core principles and challenges (see Fig. 3). Inclusive AI is multi-faceted, addressing accessibility, equity, usability, diversity, mitigating inequalities, and skills development. These facets underscore the importance of inclusive design of AI systems in healthcare, as it is important to ensure that all people have access to healthcare services and nobody will be excluded from healthcare services because of non-inclusive design of the technology or missing skills of interacting with the AI technologies. Nyariro et al. [35] recognized the

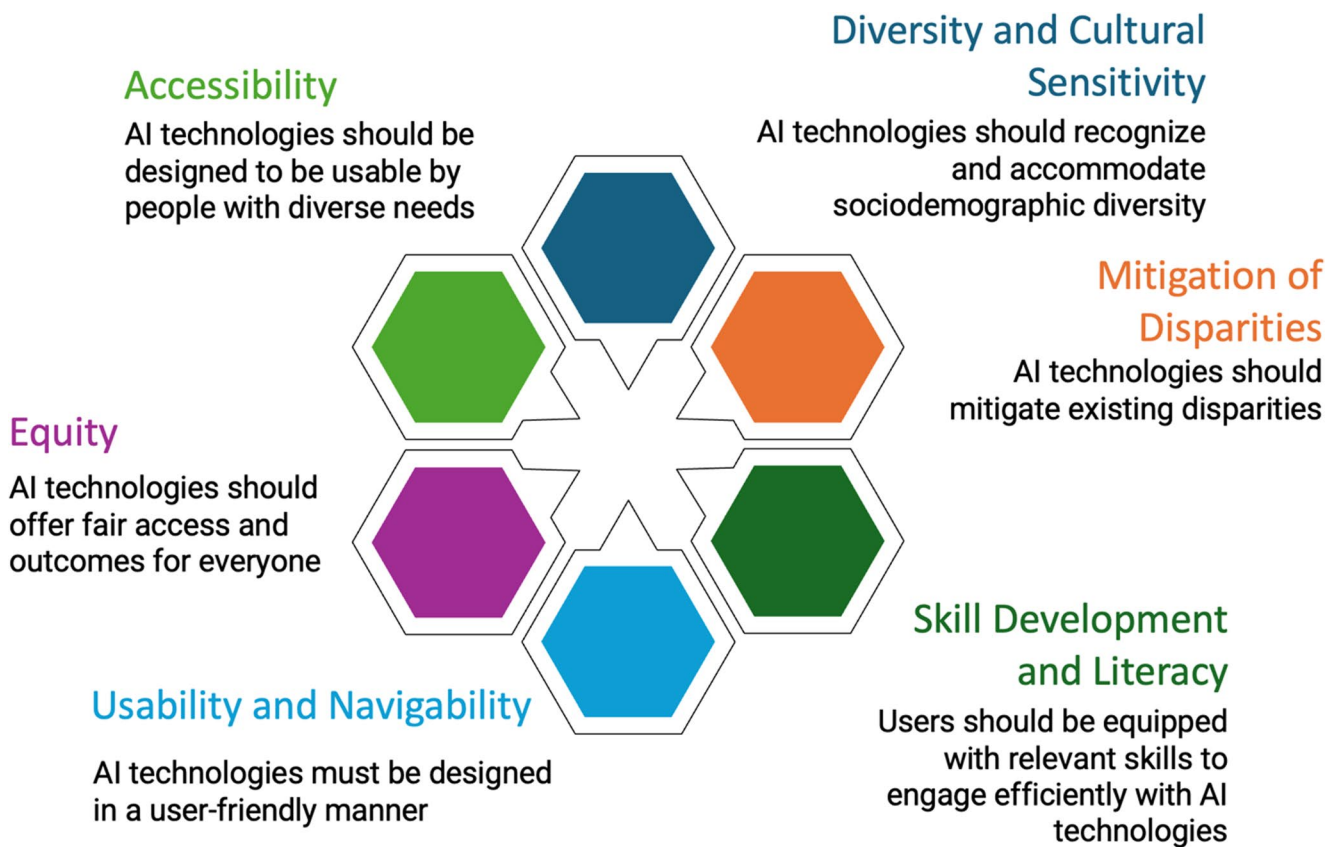


Fig. 3 Facets of inclusive AI: Inclusive AI is multifaceted, addressing accessibility, equity, usability, diversity, mitigation of disparities, and skill development

need to study to what extent equity, diversity and inclusion is integrated throughout the lifecycle of AI in healthcare with results of the review still pending [35]. The relevance of usability of digital tools in healthcare has already been perceived as relevant for mHealth solutions. This fact is supported by the availability of multiple tools for assessing usability in mHealth solutions (e.g. Mobile App Rating Scale [36], mHealth App Usability Questionnaire) [37]. Future work should study whether existing tools for usability assessment are suited or require adaptation to ensure usability in AI technologies. Reducing inequalities will only be possible if AI technologies are designed to avoid exacerbating health inequalities. While bias in AI technologies has been assessed and studied already [38], effective strategies to address and overcome biased outcomes of these systems are still lacking.

4.3 Facilitators and challenges of inclusive AI in healthcare

Key facilitators (strengths and opportunities) of inclusive AI technologies in healthcare include technological innovation and advances such as increased access to smartphones, advances in natural language processing [39], machine

learning [40], and generative AI, all of which improve accessibility and inclusivity. These technologies also contribute significantly to advancing benefits health equity by empowering patients, enabling personalized medicine, optimizing workflows, and creating opportunities for preventive and proactive care [41].

Potential challenges (weaknesses and threats) of inclusive AI technologies within the healthcare domain include the need for diverse data sets, and their potential to exacerbate inequities. The development of inclusive AI technologies relies on access to diverse datasets that reflect the complexity of real-world populations [1]. Ensuring that these data sets are representative and inclusive is crucial for developing effective and unbiased AI systems. Another challenge is cultural competence and sensitivity. Inclusive AI technologies must be designed with cultural competence and sensitivity in mind, taking into account the diverse needs and preferences of patients from different backgrounds [10]. This includes ensuring that AI-powered decision support systems are aware of cultural differences in health beliefs and practices. Further research is needed to evaluate the effectiveness and equity of inclusive AI technologies in different application areas, ensuring that these technologies are

having a positive impact on health outcomes and reducing disparities.

Findings from our review align with the “digital determinants of health” framework [27] which emphasizes how technological design, accessibility, and governance shape health equity outcomes. The threats we have identified, such as bias in algorithms, underrepresentation of marginalized groups, and unequal infrastructure, reflect the proposed key digital determinants (e.g., inequitable access to digital tools, varied digital literacy, and systemic bias embedded in technology development) [27]. Likewise, the opportunities identified in our review, such as co-design methodologies, international collaborations, and the adoption of inclusive frameworks, can provide specific approaches on how to address these determinants. Our review also places inclusive AI within the discussion of how digital factors affect health and therefore emphasizing the need to integrate socio-technical and equitable perspectives in the design, implementation and governance of AI-based technologies in healthcare. Our findings also support related approaches, such as the Social Justice framework for bridging the digital divide as suggested by Rubeis et al. [26], and the Human Rights values framework used by Israni et al. [25], highlighting the need for AI systems that are both technically robust and equity-driven.

To ensure the long-term usability and adaptability of inclusive AI systems, continuous refinement and regular updates are essential. It is crucial to engage in discussions that explore sustainable strategies for maintaining and improving inclusivity over time.

The critical appraisal of the included articles indicates that the evidence base is primarily composed of expert opinion articles that met established JBI quality criteria, suggesting a generally strong conceptual foundation but limited empirical validation. However, the predominance of expert-driven contributions highlights a gap in systematically evaluated, data-driven evidence in the field of inclusive AI in healthcare. The analysis also reveals a wide range of frameworks and guidelines that support inclusive AI in healthcare, each addressing different aspects of AI implementation. These frameworks include ethical design principles, bias mitigation strategies, institutional trust, accessibility considerations, social equity and regulatory compliance. The presence of multiple frameworks suggests a strong interdisciplinary approach but also indicates a potential need for standardization or harmonization. Several frameworks specifically focus on ensuring ethical AI development and mitigating biases. The Bias Mitigation Framework and ethical AI design principles underscore the importance of addressing algorithmic bias and fairness. Similarly, the Social Justice Framework for Bridging the Digital Divide, the Human Rights Values Framework, and

principles of distributive justice emphasize the need for equity, fairness, and the protection of marginalized communities. Some frameworks align with internationally recognized standards, such as the IEEE 7000 standard for ethical system design, the W3C Accessibility Guidelines, and technology acceptance theories.

These frameworks provide structured methodologies for AI development, ensuring that AI-driven healthcare solutions comply with global accessibility and ethical standards. Frameworks like ACCEPT-AI, SPIRIT-AI, and CONSORT-AI are tailored specifically for AI applications in clinical and healthcare settings, ensuring transparency and rigor in AI-driven medical interventions. Some frameworks focus on cultural and philosophical perspectives, such as the Ubuntu philosophy, which emphasizes the interconnectedness of individuals and their communities. The inclusion of Sustainable Development Goal 17 (Partnership for the Goals) highlights the importance of cross-sector collaboration and global cooperation in developing inclusive AI solutions. This underscores the need for partnerships between healthcare institutions, policymakers, AI developers, and community stakeholders.

Although we identified a variety of ethical and inclusive frameworks, each was mentioned only once across the reviewed articles, indicating limited adoption or awareness within the field. Frameworks like Ubuntu philosophy, used by Amugongo et al. [6], which emphasize community and interconnectedness, may be underutilized due to a lack of familiarity, limited interdisciplinary integration, or challenges in applying culturally specific philosophies within predominantly technology-driven research. Greater efforts are needed to promote and adapt diverse frameworks to enhance the inclusivity and ethical grounding of AI in healthcare. In summary this study examines the facets of inclusive AI, and identifies application areas, potential challenges, recommended strategies, solutions, and regulatory frameworks needed to realize inclusive AI technologies in fostering patient-centered healthcare.

4.4 Recommendations to achieve inclusive AI in healthcare

For the six identified facets of inclusive AI and the facilitators and challenges, we came up with strategies for the purpose of achieving inclusive AI. They are listed in Table 4.

While some strategies such as application of user-centered, participatory design principles or continuous assessment of usability hold true for other technologies such as M-Health solutions, there are some specific strategies recommended.

Accessibility guidelines such as those from the W3C for web content accessibility [42] are available. Among the first

Table 4 Strategies recommended to address the facets of inclusive AI in healthcare

Facets of inclusive AI	Stakeholders	Recommended strategies
Accessibility	AI developers Researchers End-users Healthcare providers	• Apply user-centered, participatory design involving a diverse user set • Implement accessibility features (e.g. global accessibility standards) • Apply personalization techniques
Equity	AI developers Researchers Regulatory authorities Healthcare providers	• Design AI tools that work on low-cost hardware • Use interpretable models and explainability tools to ensure transparency
Usability and navigability	Researchers End-users Healthcare providers	• Develop usability metrics for AI technologies • Study usability and navigability with a diverse set of users • Apply user-centered, participatory design
Diversity and cultural sensitivity	AI developers Data scientists Researchers Regulatory authorities	• Establish datasets that represent various genders, ethnicities, socioeconomic statuses etc. and underrepresented groups • Ensure transparency about data sources
Mitigation of disparities	AI developers Researchers Independent auditors Regulatory agencies	• Regularly auditing AI algorithms to identify and correct biases • Use fairness metrics to evaluate AI outputs across sociodemographic groups
Skills development and literacy	Healthcare institutions Universities Government health departments AI developers	• Conduct training programs for healthcare professionals and patients • Include explainability features to the AI technologies • Ensure explainability for a diverse set of users

initiatives to address accessibility in AI systems, the Canadian government stands out by providing a comprehensive technical guide aimed at fostering accessible and equitable AI technologies [43].

This pilot program is a pioneer example, offering guidelines and best practices to ensure AI systems are designed with inclusivity at their core [43]. Personalization techniques could also help in making AI systems accessible as the personalization can consider cognitive disabilities or other restrictions in content provision and interaction. Usability metrics specifically for AI technologies in healthcare are still missing.

To address diversity, and avoid disparities, datasets need to be carefully established. A regular audit process for AI

algorithms could be established to identify and correct biases. Applying fairness metrics could support this [44].

Skills and literacy can be developed by training programs and by including explainability features into AI systems. This would not only contribute to literacy, but also to trust. However, explainability features should be inclusive themselves. Tielman et al. provide a roadmap for explainable AI for people with cognitive disabilities [45]. More research in this field is needed.

4.5 Research agenda for improving health inclusion when deploying AI systems

AI-based tools are being adopted by health practitioners. In contrast, evidence from the literature suggests that most patients are still hesitant due to several factors such as privacy, data security, risk of misinformation, trust, etc [30]. The existence of these factors may decrease fairness and inclusiveness in healthcare when AI is deployed by patients [31]. Therefore, this section identifies factors that impact inclusive AI in healthcare as described below. The information or decision made by AI algorithms or systems related to patients' care is at times difficult to define in simple terms. Furthermore, information accuracy is another factor [30], prompting the capabilities of AI systems to deliver up-to-date, adaptable, and standardized patient educational information. Thus, AI systems such as LLMs should consistently provide evidence-based information [10].

Furthermore, the transparency, interpretability, explainability, and visibility of AI systems is a factor that can influence inclusion as this assess how clear AI communicates information [30]. Transparency means that the data sources, methodologies, and potential limitations of AI systems and algorithms should be specified as most AI systems are black boxes [46]. In addition, there is need for fair and unbiased AI systems [30, 31]. Fairness ensures that AI algorithms treat everyone equally [6]. In developing AI systems, sensitive information used by AI may lead to data privacy breaches [30]. Another factor is bias in responses produced by AI algorithms [10], which can substantially affect healthcare decisions, particularly for marginalized and underserved communities [29].

Moreover, the lack of participatory design approaches by developers of AI systems is also a factor that limits the actualization of inclusive AI in healthcare. Participatory design promotes health equity equality, and justice. This is because it creates an avenue to engage marginalized populations during the co-design, development, and evaluation of AI systems [1]. Participatory design of AI systems comprises of diverse stakeholders such as philosophers, computer scientists, health personnels, patients, etc [1]. Gender biases are another determinant that has impacted AI systems such

as Large Language Models (LLM). As reported in the literature most LLM show prejudice in gender when it comes to healthcare [46]. This has also result in the unjust decision making, allocation of resources, and limited opportunities for women resulting to prejudices and stereotypes for all [46].

The accessibility of AI systems such as ChatGPT can indeed help in bridging health information divide, especially in low-income communities. Although existing AI systems are not developed to diverse patient groups which results in disparity in healthcare. Thus, there is need for AI systems to develop systems that can accurately interpret and convey health related information across languages, thus facilitating understanding and trust between patients and healthcare providers [10].

The critical appraisal indicates that the evidence is primarily composed of expert opinion articles that met established JBI quality criteria, suggesting a generally strong conceptual foundation but limited empirical validation. This, together with the presence of at least one empirical study with a high risk of bias identified through PROBAST, underscores the need for stronger methodological rigor and more robust empirical evidence to validate proposed inclusive AI frameworks and recommendations in real-world healthcare settings. Future research should evaluate the effectiveness of the frameworks considered in real-world AI implementation. By identifying gaps and overlaps among existing frameworks, a more unified, comprehensive, and adaptable framework could be developed for inclusive AI in healthcare. Such standardized framework could address the weaknesses and threats identified in our analysis. It would streamline AI development, ensure fairness, improve trust, and foster global AI adoption in healthcare. It could also address critical gaps in bias mitigation, accessibility, and regulatory alignment, making AI-driven healthcare solutions more equitable and effective for all patients, especially marginalized communities. Future research should also prioritize recommendations based on their feasibility and potential impact. At short term, regular actions such as conducting bias audits, implementing transparency measures, and enhancing data privacy measures could help to identify and mitigate unfairness in AI algorithms. At medium and long term, it would be desirable to adopt more user-centered approaches involving diverse stakeholders in order to create equitable AI systems.

4.6 Limitations of this study

Several limitations should be considered when interpreting the results of this review. First, our selection of sources was mostly linked to publications that explored or contributed to “inclusive AI”. Thus, there is a potential of omitting other

relevant studies linked to patient-centered healthcare due to the terminologies used in the search strategy. Moreover, we did not include variations of these terms, additional keywords, or expand the search to the entire manuscript. As a result, relevant articles that used alternative terminology or did not explicitly include these keywords in the abstract may have been overlooked. Additionally, our search was limited to English-only databases and grey literature such as non-governmental reports on AI in low- and middle-income countries, which introduces a potential selection bias. However, the inclusion of articles published between 2020 and 2024 ensures that the review reflects the latest trends and developments in the field of inclusive AI in healthcare. Additionally, although our search was comprehensive and used several databases, it was limited to publications indexed in established databases, which may have excluded relevant or recent works published in other platforms such as preprints or institutional repositories.

Although two authors were involved in the data extraction process, the data extraction was not performed in duplicate, which may introduce potential bias or error. This increases the risk of missing subtle nuances or differing interpretations in coding the strengths, weaknesses, opportunities, and threats (SWOT), potentially affecting the accuracy of the findings.

Additionally, most included articles were expert opinion papers, which, despite meeting established quality criteria, may limit the strength of the evidence due to the lack of empirical validation. Furthermore, one empirical study assessed using PROBAST was found to have a high risk of bias, particularly in the analysis domain, which may affect the reliability and generalizability of its findings.

Moreover, the complexity of the topic and the overlap of certain concepts may have led to the omission of some relevant details or misclassifications. Further research should consider duplicate extraction to improve reliability. Moreover, issues related to the impact of AI in health care related to race and gender were not addressed in detail as this has been investigated in prior studies. While our review focused primarily on clinical AI applications, many of the identified strengths and challenges, such as bias, inclusivity, and transparency, can also be relevant to non-clinical AI domains like public health surveillance. Future research should explore how these findings translate to broader AI use cases further also considering the impact of AI use in healthcare related to race and gender.

5 Conclusions

In conclusion, the use of AI in healthcare offers significant opportunities to improve health and well-being, but its implementation must be approached with careful consideration of inclusivity, transparency and equitable use. This study highlights the importance of inclusive AI in addressing health inequalities, and highlights both the strengths and challenges associated with its use. Through an analysis of different AI applications and inclusivity frameworks, the findings emphasize the need for robust strategies to mitigate bias, improve accessibility, and promote patient-centered healthcare solutions. The SWOT analysis shows that while AI can advance health equity and innovation, it also has the potential to exacerbate existing inequalities if not properly regulated. To maximize the benefits of AI in healthcare, a participatory and inclusive approach is essential to ensure that all stakeholders - including patients, clinicians and policymakers - are involved in shaping AI-driven solutions. Going forward, the development and implementation of inclusive AI technologies must be guided by ethical principles, regulatory frameworks and interdisciplinary collaboration to create a healthcare system that is equitable, transparent and beneficial for all.

Future efforts will focus on exploring how inclusive, patient-centered AI systems can be designed in partnership with all key stakeholders in the healthcare sector and on developing a standardized framework for developing inclusive AI solutions in healthcare. Additionally, we will explore the impact of AI as related to health equity in society. Also, the identified “facets of inclusive AI” (as seen in Fig. 3) will be expanded into a conceptual model, and then the model will be evaluated using qualitative and quantitative data collected using survey questionnaire and focus group interview.

Appendix 1: Full search strategy

Online database	Search terms	Results
PubMed	inclusive AI[Title/Abstract]	13
PsycINFO	AB inclusive AI	8
CINAHL	AB inclusive AI	9
Academic Search Premier	AB inclusive AI	75
IEEE Xplore	(“Abstract”:inclusive AI)	163
Scopus	ABS (inclusive AND a.i.)	1109
TOTAL		1377

Search date: 26 September 2024

Appendix 2: Excluded Articles During Full Text Screening (N = 21) And Exclusions Reasons

No Health-Related Context (n = 5):

- de Simone, F., M. Casillo and S. Collina (2024). *AI and Inclusivity: Co-designing for Disability Empowerment*. Lecture Notes in Networks and Systems.
- Gross, N. (2023). “What ChatGPT Tells Us about Gender: A Cautionary Tale about Performativity and Gender Biases in AI.” *Social Sciences* 12(8).
- Queerina, O. O., A. Ovalle, A. Subramonian, A. Singh, C. Voelcker, D. J. Sutherland, D. Locatelli, E. Breznik, F. Klubicka, H. Yuan, J. Hetvi, H. Zhang, J. Shriram, K. Lehman, L. Soldaini, M. Sap, M. P. Deisenroth, M. L. Pacheco, M. Ryskina, M. Mundt, M. Agarwal, N. McLean, P. Xu, A. Pranav, R. Korpan, R. Ray, S. Mathew, S. Aroora, S. John, T. Anand, V. Agrawal, W. Agnew, Y. Long, Z. J. Wang, Z. Talat, A. Ghosh, N. Dennler, M. Noseworthy, S. Jha, E. Baylor, A. Joshi, N. Y. Bilenko, A. McNamara, R. Gontijo-Lopes, A. Markham, E. Dong, J. Kay, M. Saraswat, N. Vytla and L. Stark (2023). *Queer In AI: A Case Study in Community-Led Participatory AI*. ACM International Conference Proceeding Series.
- Wald, M. (2021). “AI Data-Driven Personalisation and Disability Inclusion.” *Frontiers in Artificial Intelligence* 3.
- Zhuang, K. V. and G. Goggin (2024). “New possibilities or problems for disability and inclusion? The case of AI and ADMs across work.” *Telematics and Informatics* 92.

No Artificial Intelligence Related Context (n = 2):

- Brooks, A. L. (2023). *Technologies for Inclusive Well-being: IT and Transdisciplinary Applications*. Lecture Notes in Networks and Systems.
- Crouch MTC, Andrew NSST. Strategies for inclusivity of American Indian and Alaska Native peoples in behavior therapy research: Within-group diversity, data, and ethical recommendations. *the Behavior Therapist*. 2022;45(5):168–74.

No Topic of Inclusion (n = 2):

- Padmapriya, M., S. Pasupathy, R. Sumathi and V. Punitha (2022). “A deep learning model framework for diabetic retinopathy detection.” *International Journal of Networking and Virtual Organisations* 27(2): 107–124.
- Joshi G, Jain A, Araveeti SR, Adhikari S, Garg H, Bhandari M. FDA-Approved Artificial Intelligence and Machine Learning (AI/ML)-Enabled Medical Devices: An Updated Landscape. *Electronics* (Switzerland). 2024;13(3).

Book, Editorial, Letters to Editor... ($n=4$).

Anshari, M., M. N. Almunawar and P. O. de Pablos (2024). *Inclusivity and accessibility in digital health*. Cavallo, J. J. and M. A. Davis (2024). “Establishing robust governance of clinical artificial intelligence software – Why radiologists should lead.” *Clinical Imaging* **110**: N.PAG-N.PAG.

Goel, R., S. K. Baral and R. Venkatesh (2022). *Artificial Intelligence and Digital Diversity Inclusiveness in Corporate Restructuring*.

Samaan, J. S., Y. H. Yeo, N. Rajeev, W. H. Ng, N. Srinivasan and K. Samakar (2024). “Inclusive AI in Healthcare: Enhancing Bariatric Surgery Education for Diverse Patient Populations.” *Obes Surg* **34**(1): 270–271.

Review Papers ($n=6$):

El Morr C, Singh D, Sawhney V, Fernandes S, El-Lahib Y, Gorman R. Exploring the Intersection of AI and Inclusive Design for People with Disabilities. *Studies in health technology and informatics*. 2024;316:556–9.

Girard, A., D. Zowghi, M. Bano and M. A. Riziou (2024). *Inclusive and Explainable AI Systems: A Systematic Literature Review*. Proceedings of the Annual Hawaii International Conference on System Sciences.

Marvin, G., N. Hellen and J. Nakatumba-Nabende (2023). *Research Paradigms for Health Equity in Intelligent Mobile Healthcare Technologies: A Critical Review*. Lecture Notes in Networks and Systems.

Pham, Q., A. Gamble, J. Hearn and J. A. Cafazzo (2021). “The need for ethnoracial equity in artificial intelligence for diabetes management: Review and recommendations.” *Journal of Medical Internet Research* **23**(2).

Rodríguez-Rodríguez, I., J. V. Rodríguez, N. Shirvanzadeh, (A) Ortiz and D. J. Pardo-Quiles (2021). “Applications of artificial intelligence, machine learning, big data and the internet of things to the COVID-19 pandemic: A scientometric review using text mining.” *International Journal of Environmental Research and Public Health* **18**(16).

Taleghani, S., (B) Kundi, F. Mobeen, Y. El-Lahib, R. Gorman and (C) El Morr (2024). *Advantages and Challenges of Using AI for People with Disabilities*. International Conference on Information and Communication Technologies for Ageing Well and e-Health, ICT4AWE - Proceedings.

No Access to Full Text ($n=2$):

Anshari M, De Pablos PO, Almunawar MN. AI-powered digital health: bridging the gap for elderly populations in the evolving healthcare ecosystem. *Digital Healthcare in Asia and Gulf Region for Healthy Aging and*

*More Inclusive Societies: Shaping Digital Future*2024. p. 265–81.

Morcos M, Duggan J, Young J, Lipa SA. Artificial Intelligence Portrayals in Orthopaedic Surgery: An Analysis of Gender and Racial Diversity Using Text-to-Image Generators. *Journal of Bone and Joint Surgery*. 2024.

Appendix 3: PRISMA

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Title			
Title	1	Identify the report as a systematic review.	N/A
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2–3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	2–3
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3–4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	3–4

Section and Topic	Item #	Checklist item	Location where item is reported	Section and Topic	Item #	Checklist item	Location where item is reported
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4	Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	4		13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4		13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	NA
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	NA		13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA		13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
					13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
				Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
				Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
				Results			
				Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	4–5
					16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	4 and Appendix 2

Section and Topic	Item #	Checklist item	Location where item is reported
Study characteristics	17	Cite each included study and present its characteristics.	5–10
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	NA
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	NA
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NA
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	12–15
	23b	Discuss any limitations of the evidence included in the review.	15
	23c	Discuss any limitations of the review processes used.	15
	23d	Discuss implications of the results for practice, policy, and future research.	xxx
Other information			

Section and Topic	Item #	Checklist item	Location where item is reported
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	NA
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	15
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	16
Competing interests	26	Declare any competing interests of review authors.	16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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Declarations

Ethics and Consent to Participate declarations not applicable.

Competing interests We confirm that there are no relevant financial or non-financial competing interests to report.

Clinical Trial Number not applicable.

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